SAKTHI ARAVIND K

Generative AI Developer | LLMs | Machine Learning | Python | RAG | Prompt Engineering

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SUMMARY

Generative AI and Machine Learning Developer with hands-on experience in building, training, and evaluating machine learning models using Python, Scikit-learn, TensorFlow, Pandas, and NumPy. Experienced in developing Retrieval-Augmented Generation (RAG) applications using LangChain, Hugging Face Transformers, OpenAI API, and vector databases such as ChromaDB. Skilled in data preprocessing, exploratory data analysis, predictive modeling, and REST API integration. Passionate about building AI-powered applications and continuously learning emerging AI technologies.

TECHNICAL SKILLS

Generative AI: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, OpenAI API, Embeddings, ChromaDB, FAISS, Chatbot Development

Machine Learning: Classification, Regression, Clustering, Feature Engineering, Model Training, Model Evaluation, Deep Learning Fundamentals

Programming: Python

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, Streamlit

Data Analysis: Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization, Basic Statistics

Databases: MySQL, ChromaDB

Web Technologies: Flask, REST APIs, Firebase

Tools: Git, GitHub, GitHub Pages

EXPERIENCE

Data Analyst & Data Science Intern Nov 2024 – Jul 2025
KGISL Micro College, Coimbatore, Tamil Nadu

- Developed machine learning models using Python, Scikit-learn, and TensorFlow for classification and regression tasks.
- Performed data preprocessing, feature engineering, model training, validation, and evaluation using Accuracy, RMSE, and R^2 metrics.
- Conducted exploratory data analysis (EDA) using Pandas and NumPy to identify patterns and generate actionable insights.
- Automated preprocessing and reporting workflows using Python while maintaining clean, reusable, and well-documented code.
- Built proof-of-concept applications using LLM APIs and prompt engineering for document summarization and question-answering use cases.
- Collaborated with mentors and peers to solve real-world data analytics problems.

PROJECTS

AI Chatbot / Document Q&A Assistant (Generative AI) | Python, LangChain, OpenAI API, ChromaDB, Streamlit

- Developed a Retrieval-Augmented Generation (RAG) application that answers questions from uploaded PDF documents using embeddings and ChromaDB.
- Implemented document chunking, vector embeddings, similarity search, and LLM-powered response generation using LangChain and OpenAI API.
- Designed prompt templates to improve answer accuracy and reduce hallucinations.
- Built an interactive Streamlit interface for document upload and conversational question answering.

Laptop Price Prediction | Python, Scikit-learn, Pandas, NumPy

- Built an end-to-end regression model involving data preprocessing, feature engineering, and hyperparameter tuning.
- Achieved an **R^2 score of 0.89** using optimized regression algorithms.

- Compared multiple regression models using cross-validation and evaluation metrics.

Customer Segmentation System | Python, K-Means Clustering

- Developed a customer segmentation model using the K-Means clustering algorithm.
- Built a preprocessing pipeline including feature scaling, categorical encoding, and outlier handling.
- Determined the optimal number of clusters using the Elbow Method and Silhouette Score.

EDUCATION

Master of Commerce with Computer Applications 2022 – 2024
PSG College of Arts and Science, Coimbatore
CGPA: 8.4

CERTIFICATIONS

- Data Science & AI/ML Program — KGISL Micro College (Nov 2024 – Jul 2025)